



Raikoke Volcano

Astronauts onboard the ISS snapped a picture of the eruption of the Raikoke volcano in Russia. <https://digit.in/july19-56>



First polarised radio signal

Astronauts have captured the first polarised radio waves from a gamma-ray burst using NASA tech. <https://digit.in/july19-57>

to be sent to, and we certainly don't want a black hole to get any closer to us either...

NAKED SINGULARITIES

Imagine a singularity that has a given amount of mass in a nearly infinitely small space, but at the same time does not have an event horizon like the ones present in black holes, then that object would be called a naked singularity. This is of course a theoretical object, but it does pop out from the extremely advanced math.

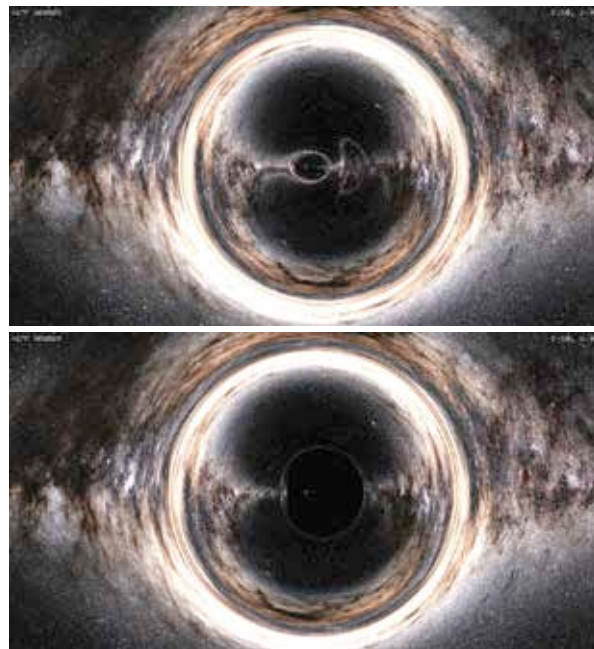
Essentially, if a singularity were to rotate faster than a given threshold speed, it would become a sort of ring shaped object, which would expose the singularity directly to normal spacetime, and thus allow observation of it. All that we could possibly learn from observing a naked singularity is what happens to an object when it is swallowed by a singularity, instead of losing sight of it at the event horizon. If the movie *Interstellar* were to have used a naked singularity, it would be very easy for the protagonists to gather data about the singularity, and no one would need to jump into it.

DIMENSIONLESS CONSTANTS

This is a weird problem that we have known for a long time. As we have observe and discover more about our universe, we come across these weird figures and numbers that have no explanation. Some of them are just pure physical constants, such as the speed of light. However, even stranger than the speed of light, and other physical constants are dimensionless constants, and here's why.

As the name suggests, these constants do not have a property, they're just a number. For example, the speed of light is a speed, and thus has the 'metres per second' description attached to the number. In contrast, the cosmological constant, has a value of 10^{-122} and is just a number. In fact, it's usually denoted by the letter Λ (lambda) in equations, because we don't want to have to write 10^{-122} all the time. The math needed to arrive

at this is beyond our understanding, but simplified, it is a constant that Einstein came up with to balance the effects of gravity in order to arrive at a static universe. Later, when Edwin Hubble found that the universe was expanding, Einstein dropped the idea of a constant from his equations. Later still, when it was discovered that the universe was not just expanding, but the expansion was accelerating (thus the further away galaxies were



Artist's impression of a naked singularity (top) vs a black hole (bottom)

from us the faster they were moving away) the cosmological constant had to be brought back in order to balance the equation again. Because we use the term dark energy to describe the thing that's causing the expansion of spacetime, the cosmological constant obviously describes the effects of dark energy. Not that we know what dark energy is to know why it causes the expansion... however, since we can observe the expansion, we can calculate the number, even if we do not know why or how it all works.

Similarly, there are more dimensionless constants that arise in our study of physics, and all of them, such as the fine structure constant, the coupling constant, proton to electron

mass ratio, and more are a complete mystery.. None of these have been explained as to why they are this specific number, we merely know it from observation and experimentation.

WHY DO WE HAVE SO MUCH WATER?

This one is strange, because if you search the internet, you will find sites proclaiming that the mystery has been solved. It's because a new

hypothesis about where all this water came from has been released. Of course the said hypothesis is based on a very niche observation and can explain some amount of the water, for example, theories about how water could have been squeezed out of rocks when the earth was still molten, have been touted as the answer before, only to later learn that they do not explain away all of the water, or even an appreciable fraction of it all. Then there's the idea of a

comet bombardment, which is another theory that has its problems. None of the theories are able to explain the vast amounts of water we have.

The reason that we are searching for an explanation is because it is thought that it would be impossible for life to get started and evolve on earth if it wasn't for all the water we have. In our search for life out there, we obviously need to search for planets that have a similar amount of water as us, and to do so, we need to know how the water got on here. As of yet, though, there's no clear explanation.

There are a few likely contenders, based on experiments, and one of those is the theory that a lot of our water originated in the asteroid belt. Be-